

Place UZ Logo from Library

Revision

Serial

Serial

Serial #

Rev03

DC-link-Caps

DC-link

Connectors

Phase_Measurements

Repeat(PhaseMeasurements,1,3)

X1
DigitalCarrierJack

OC_DC

OC_DC

OC_P[1..3]

OC_P[1..3]

OC_P

PWM_EN

PWM_H[1..3]

PWM_L[1..3]

PWM_H

PWM_L

Halfbridges

Repeat(Halfbridge,1,3)

Repeat(PHASE_)

OT_H[1..3]

OT_L[1..3]

TEMP_H[1..3]

TEMP_L[1..3]

OT_H[1..3]

OT_L[1..3]

TEMP_H[1..3]

TEMP_L[1..3]

OT_H

OT_L

TEMP_H

TEMP_L

Temp_Measurements

Repeat(TempMeasurements,1,3)

Repeat(OT_H)

Repeat(OT_L)

Repeat(TEMP_H)

Repeat(TEMP_L)

PHASE_

Repeat(I_OUT_P)

Repeat(I_OUT_N)

Repeat(U_OUT_P)

Repeat(U_OUT_N)

Repeat(PHASE_CONNECTOR_)

Repeat(OC_P)

Repeat(PHASE_)

INFO

Project
ProjectRevision
AuthorParam
ProjectDate

Design Information

LOGO1



UZ Logo

Title TopSheet.SchDoc

Revision: Rev03

Design Engineer: D. Hufnagel / A. Geiger

Project: UZ_D_Inverter.PrjPCB

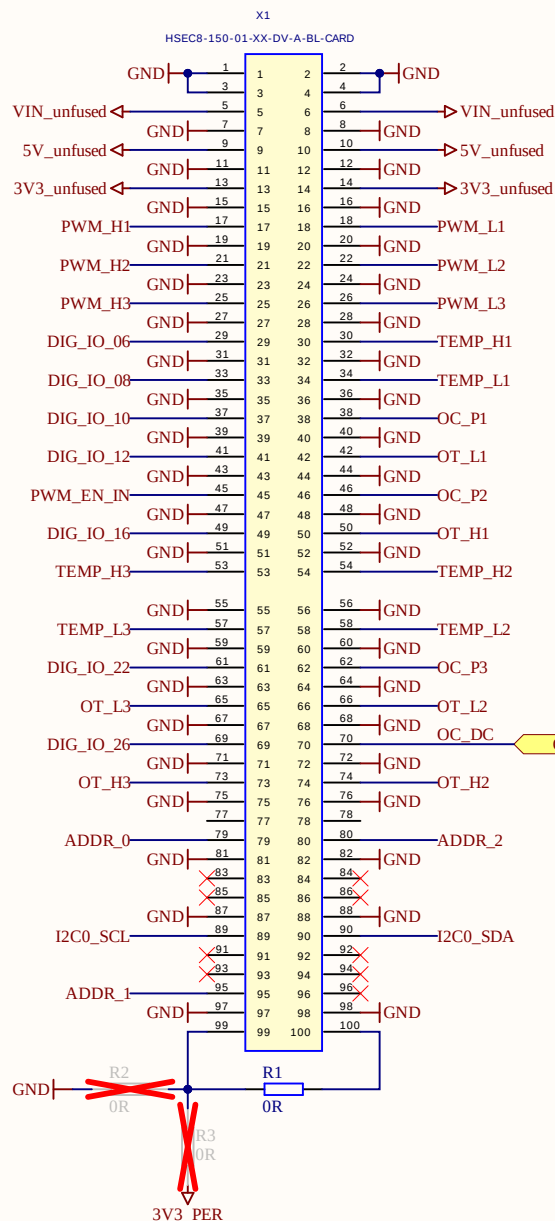
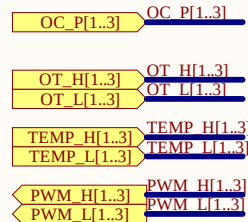
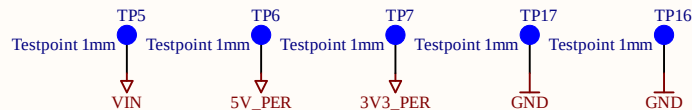
UltraZohm

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Date: 07.03.2024

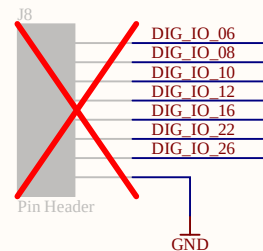
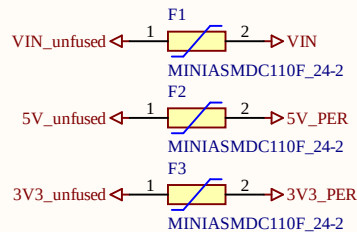
Sheet 1 of 14



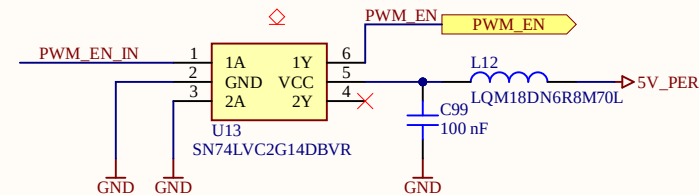


Temp_H/L signals:
Pulswidth modulated signal to the SoC. Pulswidth linear correlated to the temperature of the Mosfets
Frequency 11kHz

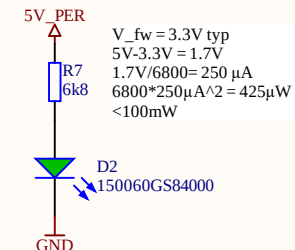
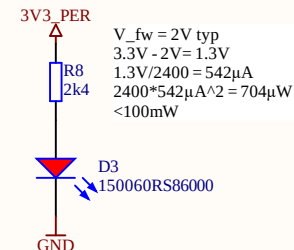
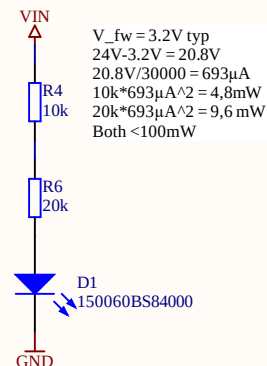
OT_H/L signals:
Active low signal.
When in normal operation, signal is 3V3.
When the measured temp exceeds 105°C, the OT signal will be set to LOW.



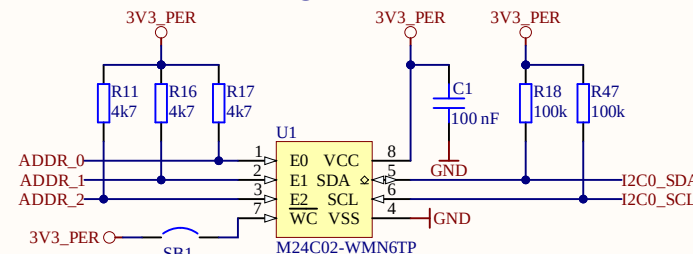
Inverting Schmitt Trigger



Status LEDs for UZ-supply rails



EEPROM



Note: place solder bump to protect EEPROM memory from write access.

Title DigitalCarrierJack.SchDoc

Revision: Rev03

Design Engineer: D. Hufnagel / A. Geiger

Project: UZ_D_Inverter.PrjPCB

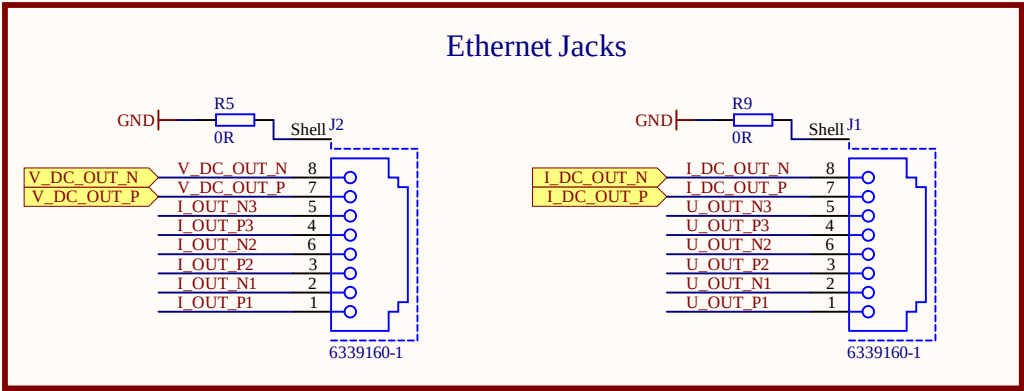
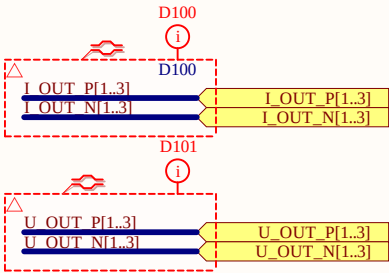
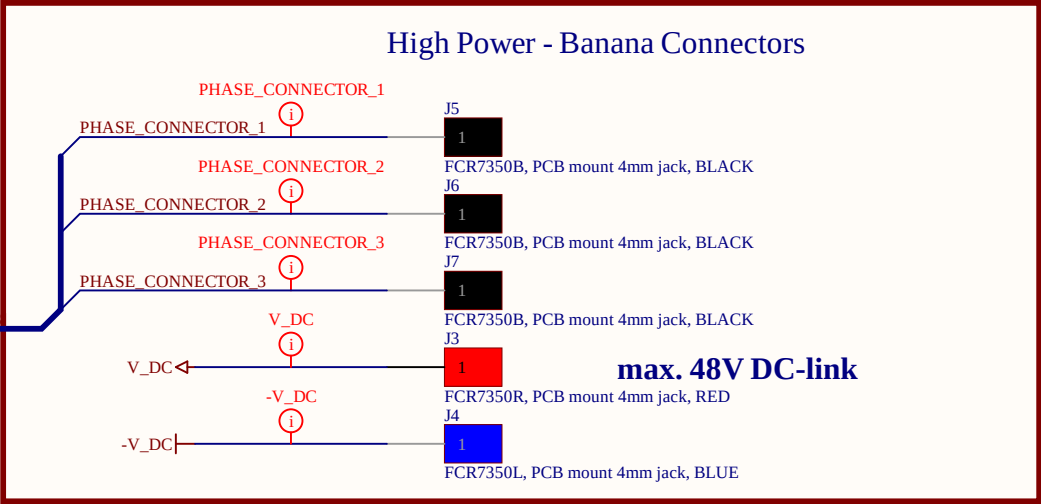
UltraZohm

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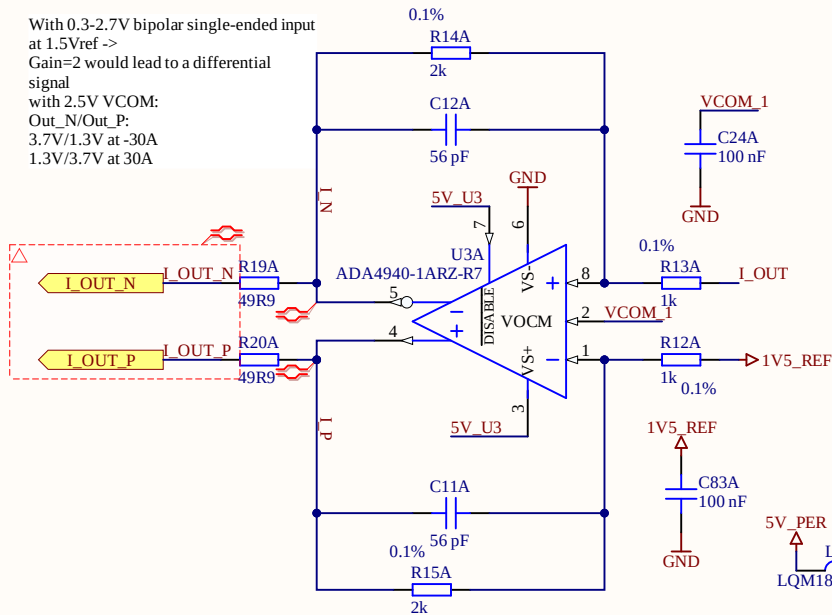
Date: 07.03.2024

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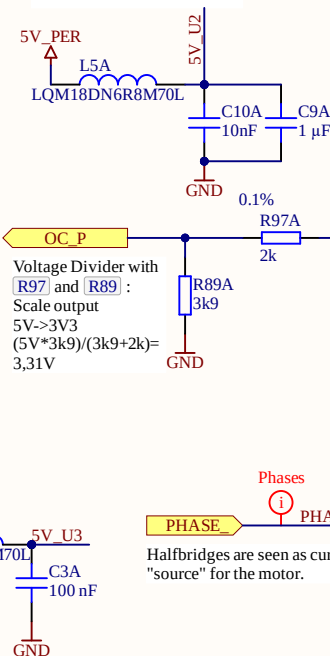




With 0.3-2.7V bipolar single-ended input at 1.5Vref ->
Gain=2 would lead to a differential signal with 2.5V VCOM:
Out_N/Out_P:
3.7V/1.3V at -30A
1.3V/3.7V at 30A

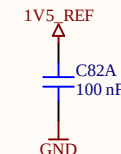


Place close to U2 OPAMP

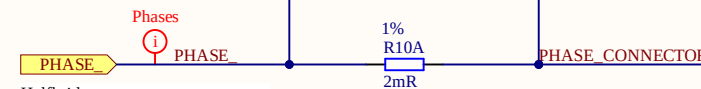


Phase Current Measurement + Single->Diff OPAMP

30A max @2mOhm = 60mV
With fixed 20V/V Gain and 1.5V ref ->
0.3 - 2.7V bipolar single-ended

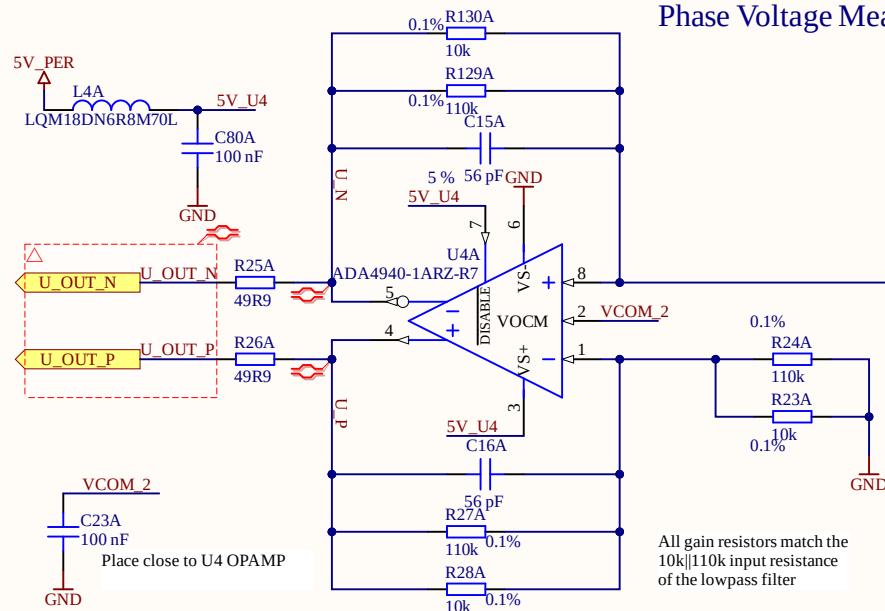


CIP sets the trigger voltage for OCP. If OUT > (VDD-CIP) || OUT < (CIP) the OCP triggers. In an OCP event, COP is set to low. With voltage divider [R94] and [R127]
 $CIP = 1k / (1k + 3k9) * 1.5V = 0.306V$
-> This translates to 29.847A



Halfbridges are seen as current "source" for the motor.

Phase Voltage Measurement + Single->Diff OPAMP



Voltage measurement scaled with voltage divider [R22] and [R21]:
Max 48V (LowPass max 60V as buffer)
 $10k || 110k \rightarrow$
 $48V * 10k / (10k + 110k) = 4V$
 $48V / (10k || 110k) = 5.2mA$

Max operation 60V:
 $60V * 10k / (10k + 110k) = 5V$
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PHASE_CONNECTOR
Direction towards the connectors and motor

0.1% LowPass Filter:
Max Target:
6000rpm at 18 pole pairs
-> $f_g = 1800Hz$
With LowPass & Feedback Lowpass. Resulting frequency response calculated via LTSpice Simulation -> 1.745kHz

All gain resistors match the $10k || 110k$ input resistance of the lowpass filter

Title Phase_Measurements.SchDoc

Revision: Rev03 Design Engineer: D. Hufnagel / A. Geiger

Project: UZ_D_Inverter.PrjPCB

UltraZohm

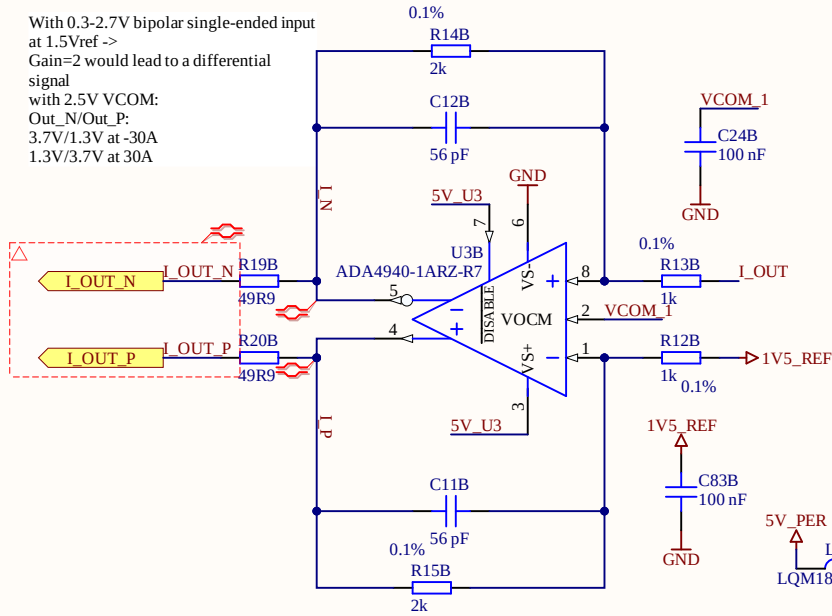
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Date: 07.03.2024

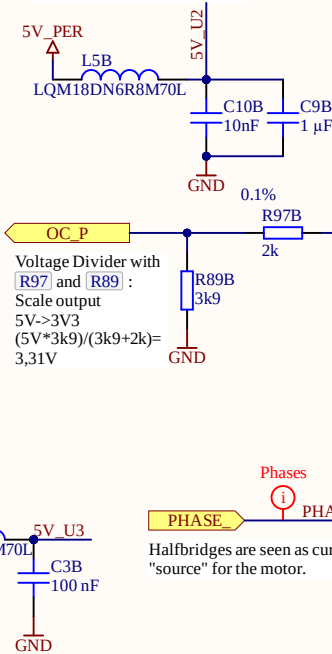
Sheet 4.1 of 14



With 0.3-2.7V bipolar single-ended input at 1.5V_{ref} ->
Gain=2 would lead to a differential signal with 2.5V VCOM:
Out_N/Out_P:
3.7V/1.3V at -30A
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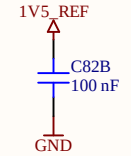


Place close to U2 OPAMP

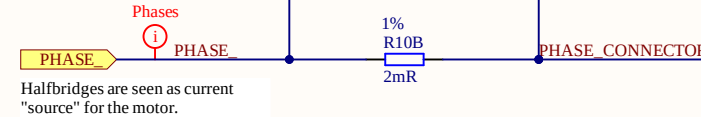


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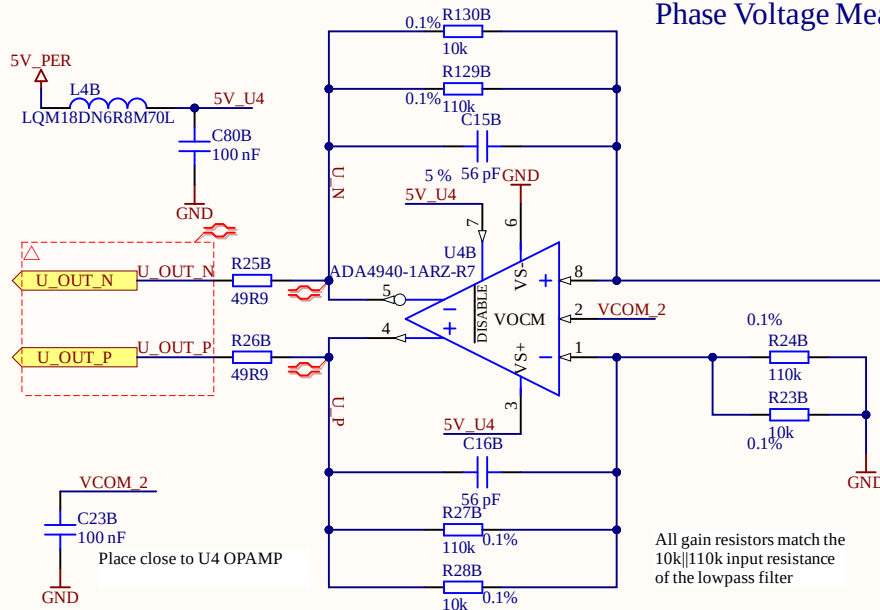


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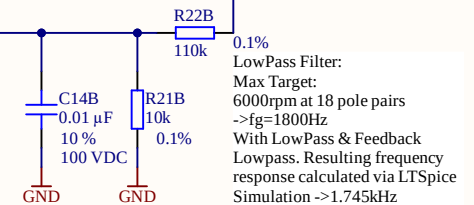
Phase Voltage Measurement + Single->Diff OPAMP



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Direction towards the connectors and motor



All gain resistors match the 10k||110k input resistance of the lowpass filter

Title Phase_Measurements.SchDoc

Revision: Rev03 Design Engineer: D. Hufnagel / A. Geiger

Project: UZ_D_Inverter.PrjPCB

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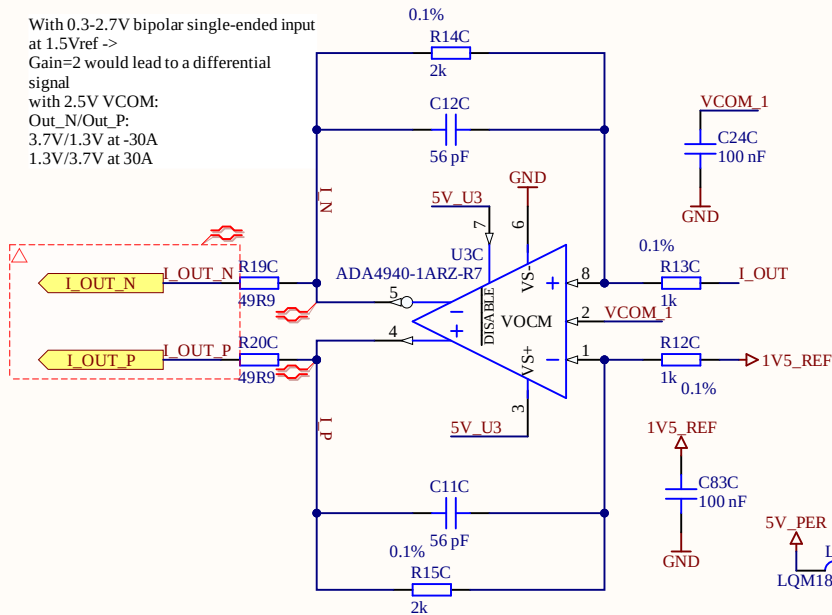
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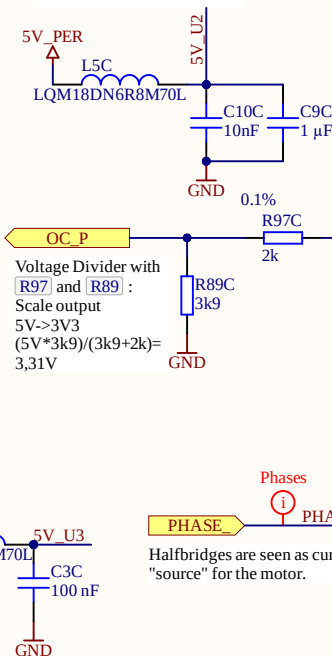
Sheet 4.2 of 14



With 0.3-2.7V bipolar single-ended input at 1.5Vref ->
Gain=2 would lead to a differential signal
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3.7V/1.3V at -30A
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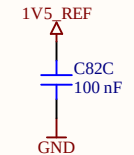


Place close to U2 OPAMP

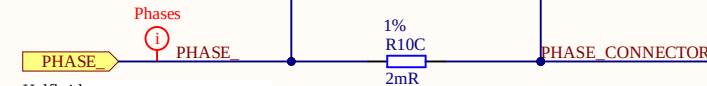


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With fixed 20V/V Gain and
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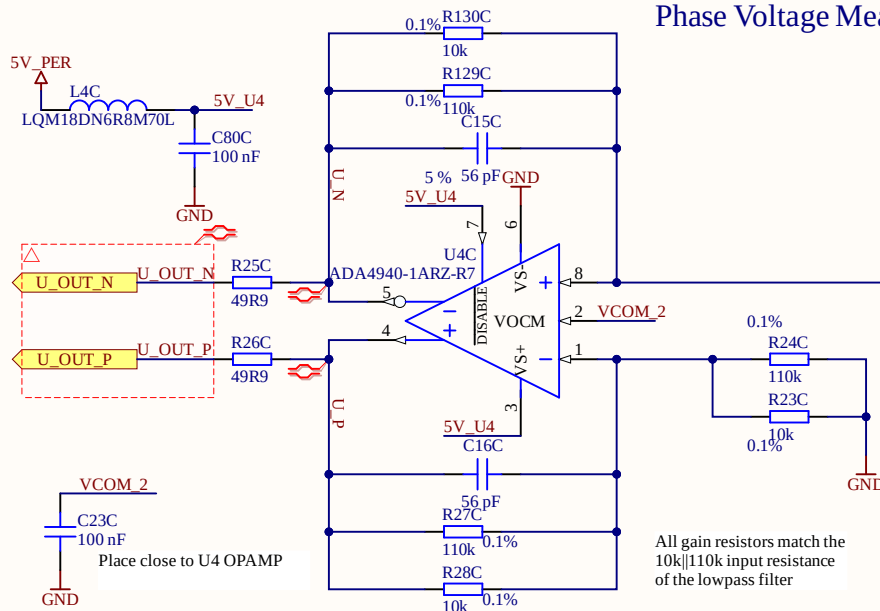


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Max Target:
6000rpm at 18 pole pairs
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With LowPass & Feedback
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response calculated via LTSpice
Simulation -> 1.745kHz

All gain resistors match the $10k || 110k$ input resistance of the lowpass filter

Title Phase_Measurements.SchDoc

Revision: Rev03 Design Engineer: D. Hufnagel / A. Geiger

Project: UZ_D_Inverter.PrjPCB

UltraZohm

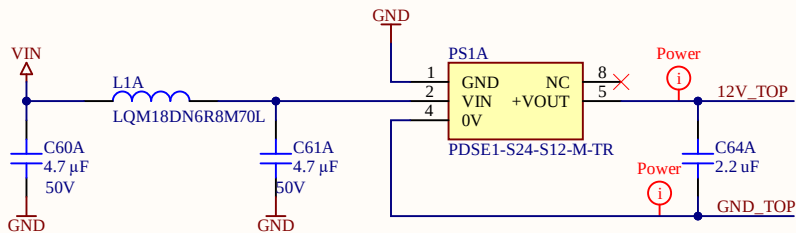
www.ultrazohm.com

Date: 07.03.2024

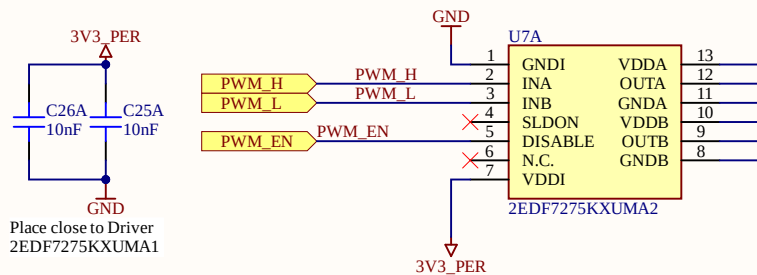
Sheet 4.3 of 14



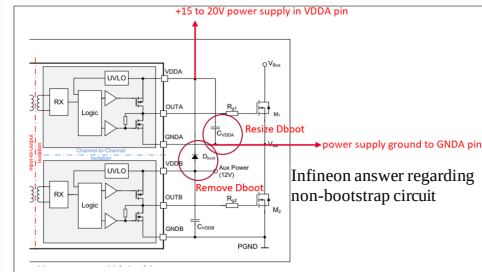
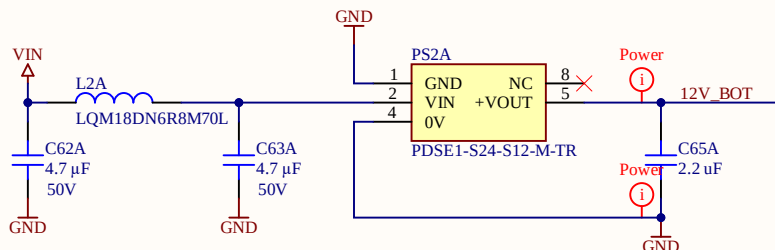
Isolated Top Side Gate Supply



Gate Driver



Isolated Bottom Side Gate Supply

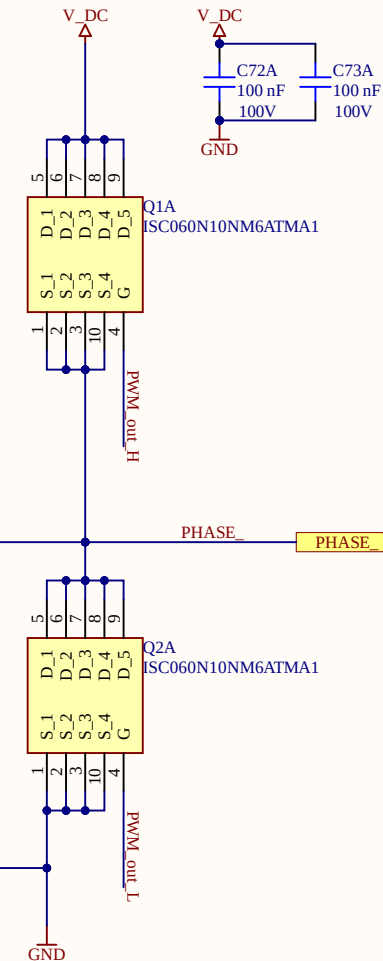


IG_charging_peak <4A
IG_discharging_peak <8A

->Max currents:
IG_charging = $12V / (47 + 0.42) \Omega = 253mA$
IG_discharging = $12V / (47 + 0.18) \Omega = 254mA$

Datasheet said minimum
20x CISS for
output-caps:
 $2 \times 100nF > 20 \times 2.5nF$

MOSFETS



Title Halfbridges.SchDoc

Revision: Rev03

Design Engineer: D. Hufnagel / A. Geiger

Project: UZ_D_Inverter.PrjPCB

UltraZohm

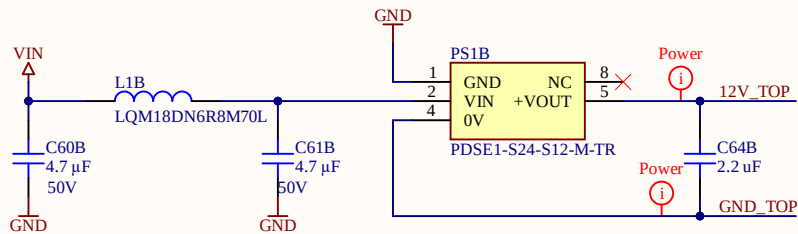
www.ultrazohm.com

Date: 07.03.2024

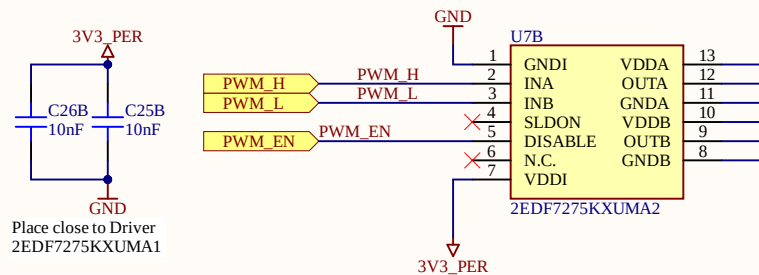
Sheet 5.1 of 14



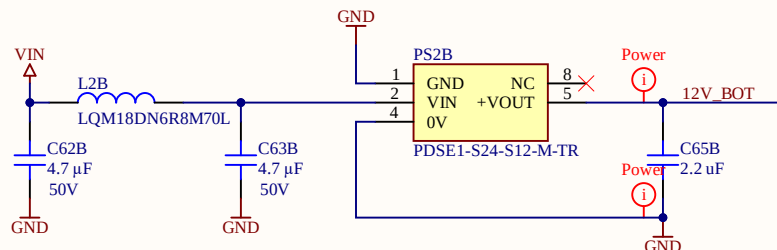
Isolated Top Side Gate Supply



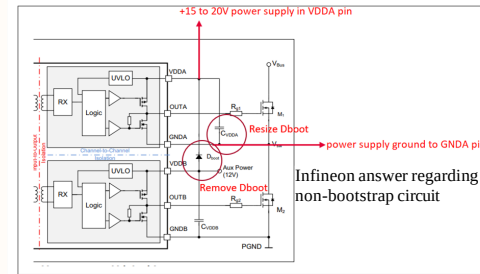
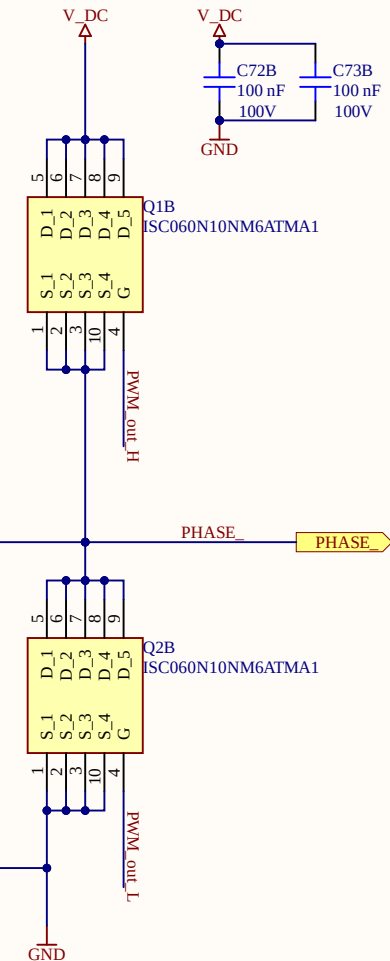
Gate Driver



Isolated Bottom Side Gate Supply



MOSFETS



IG_charging_peak <4A
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->Max currents:
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2x100nF > 20x2.5nF

Title Halfbridges.SchDoc

Revision: Rev03

Design Engineer: D. Hufnagel / A. Geiger

Project: UZ_D_Inverter.PrjPCB

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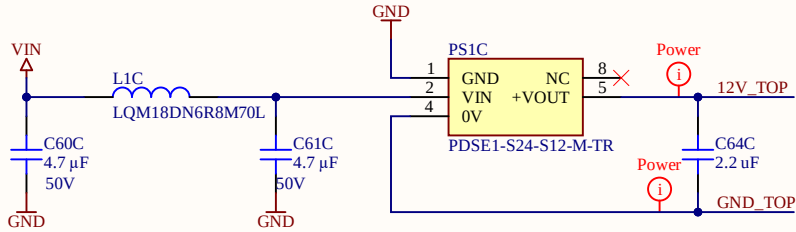
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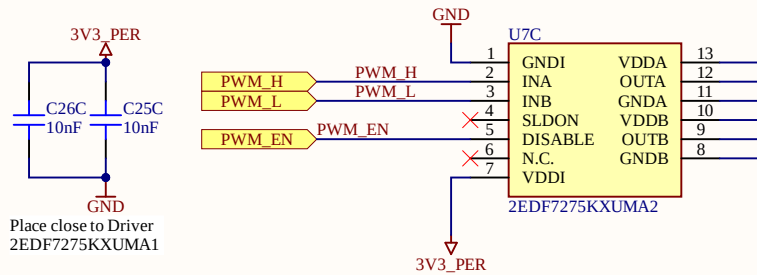
Sheet 5.2 of 14



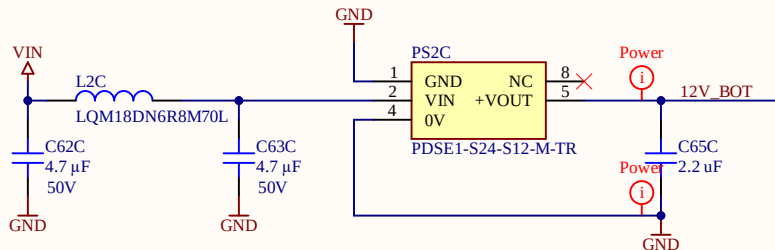
Isolated Top Side Gate Supply



Gate Driver



Isolated Bottom Side Gate Supply

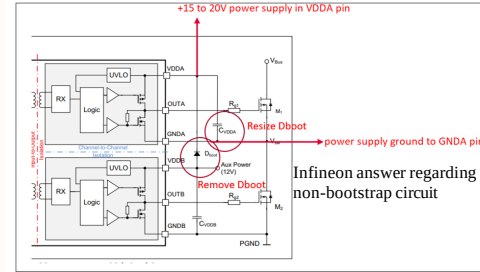
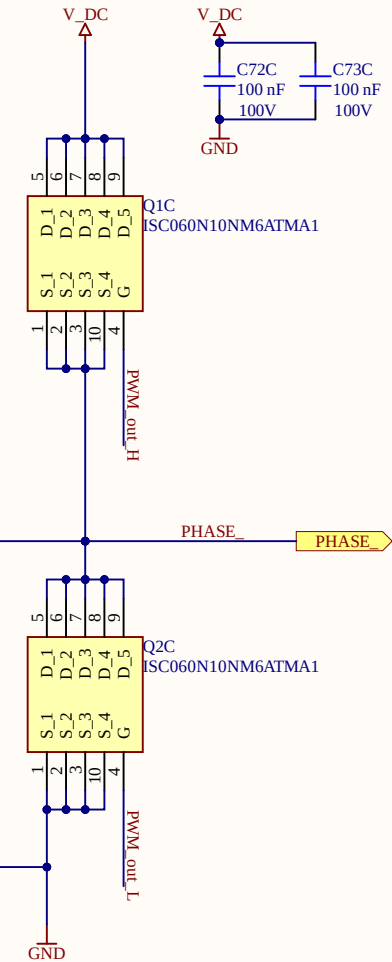


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MOSFETS



Title Halfbridges.SchDoc

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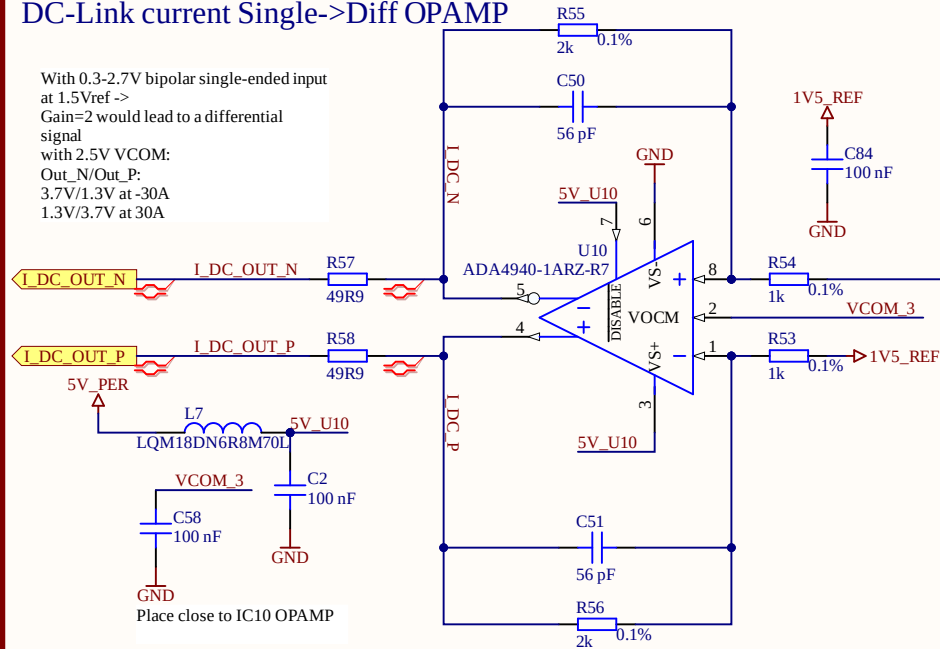
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Sheet 5.3 of 14



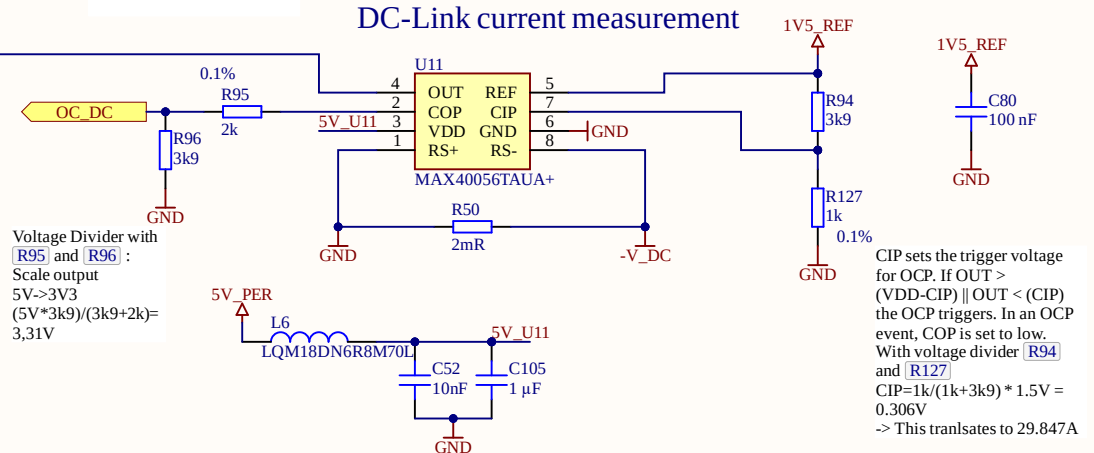
DC-Link current Single->Diff OPAMP

With 0.3-2.7V bipolar single-ended input at 1.5Vref ->
Gain=2 would lead to a differential signal with 2.5V VCOM:
Out_N/Out_P:
3.7V/1.3V at -30A
1.3V/3.7V at 30A



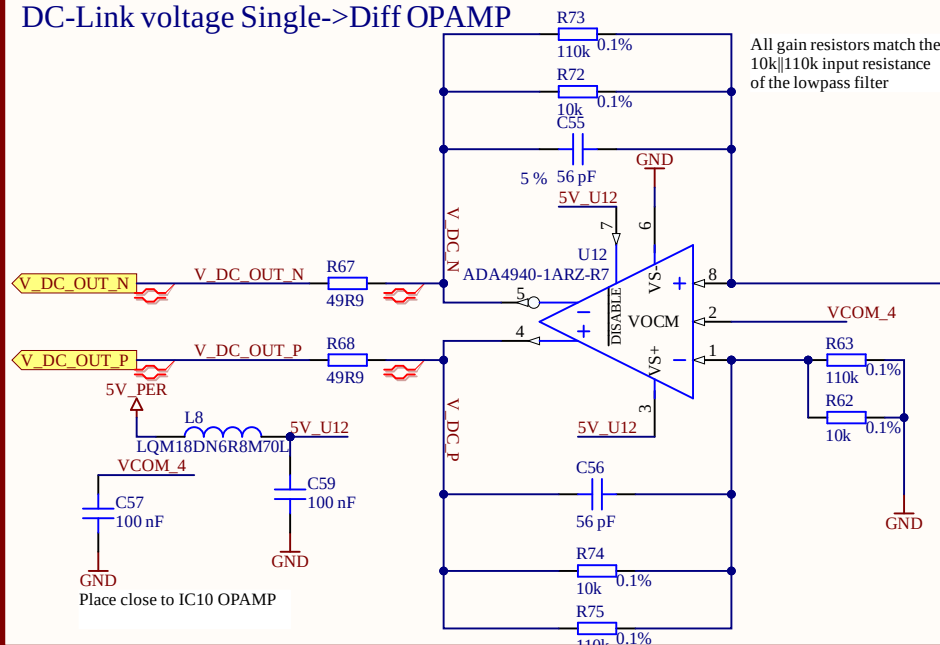
30A max @2mOhm = 60mV
With fixed 20V/V Gain and 1.5V ref ->
0.3 - 2.7V bipolar single-ended

DC-Link current measurement



DC-Link voltage Single->Diff OPAMP

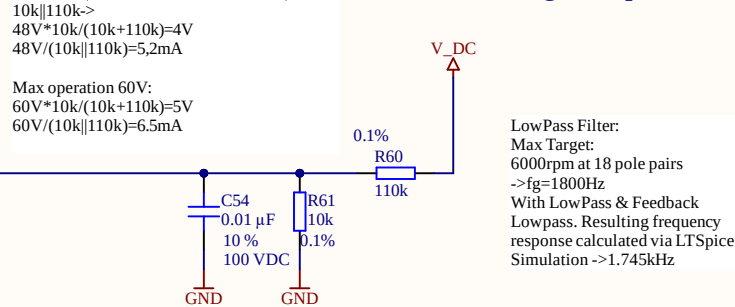
All gain resistors match the $10k \parallel 110k$ input resistance of the lowpass filter



Voltage measurement scaled with voltage divider **R60** and **R61** :
Max 48V (LowPass max 60V as buffer)
 $10k \parallel 110k \rightarrow$
 $48V \cdot 10k / (10k + 110k) = 4V$
 $48V / (10k \parallel 110k) = 5.2mA$

Max operation 60V:
 $60V \cdot 10k / (10k + 110k) = 5V$
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DC-Link Voltage lowpass



Title DC-link.SchDoc

Revision: Rev03

Design Engineer: D. Hufnagel / A. Geiger

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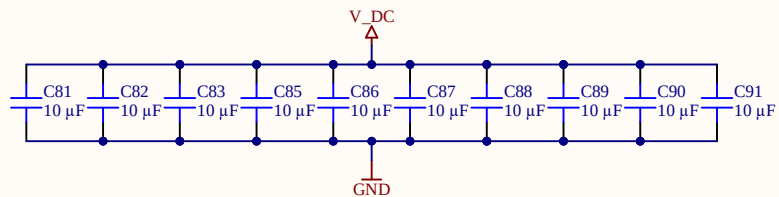
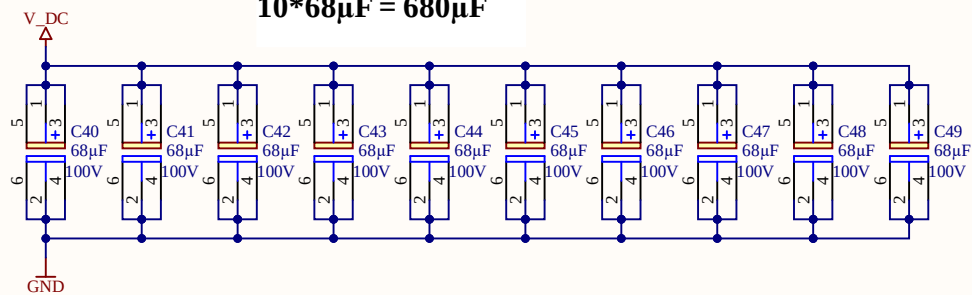
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Date: 07.03.2024

Sheet 6 of 14



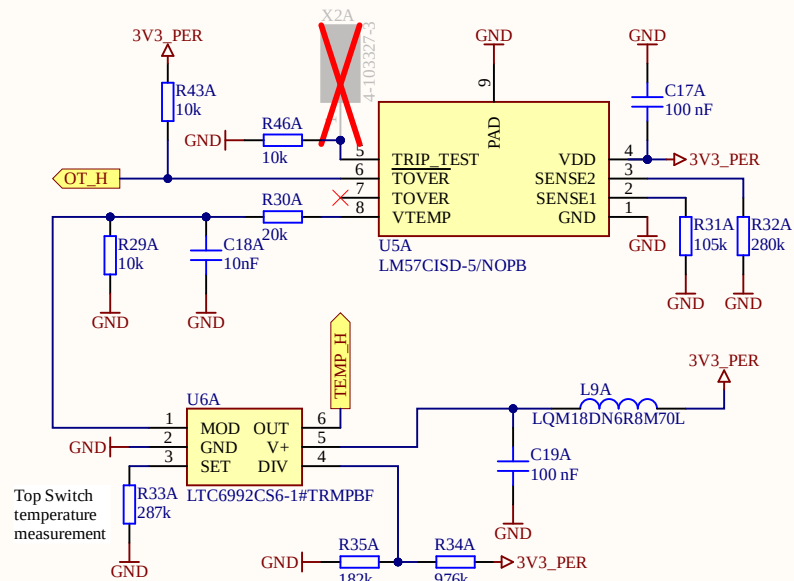


Additional MLCC caps with low internal R
 $10 \times 10\mu\text{F} = 100\mu\text{F}$

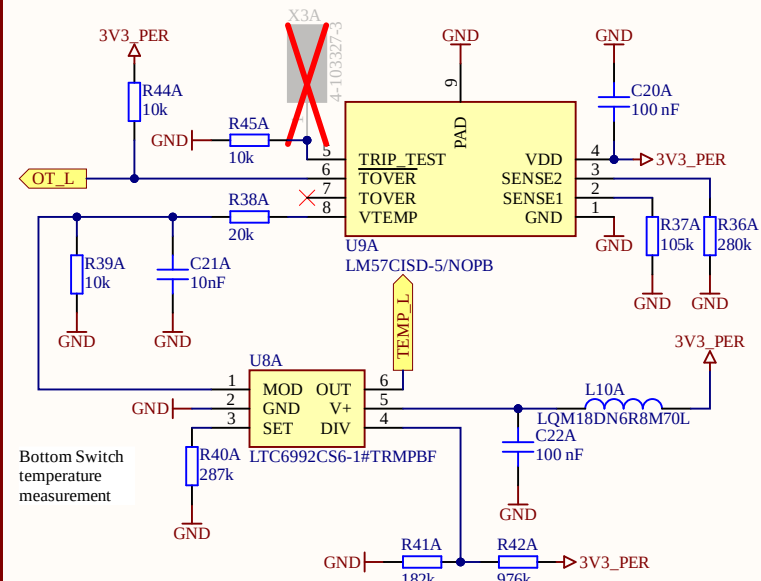
Total DC-Link capacitance
 $470\mu\text{F} + 100\mu\text{F} = 570\mu\text{F}$



Top MOSFET Temp Measurement

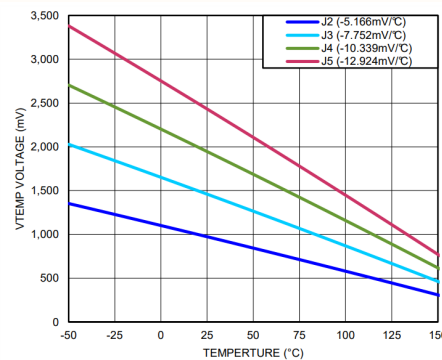


Bottom MOSFET Temp Measurement



$R_SENSE1 = \left(\frac{R31}{R37} \right) = 105k$
 $R_SENSE2 = \left(\frac{R32}{R36} \right) = 280k$
 is Gain
 -> J4 -> Leads to:
 1.11V@105°C
 2.00V@20°C

VTEMP has voltage divider with 20k/10k:
 MOD input voltages->
 $1.11V * 10 / (10 + 20) = 0.37V @ 105^\circ C$
 $2.00V * 10 / (10 + 20) = 0.67V @ 20^\circ C$
 MOD input should be between 0.14V and 0.9V



Title Temp_Measurements.SchDoc

Revision: Rev03 Design Engineer: D. Hufnagel / A. Geiger

Project: UZ_D_Inverter.PrjPCB

UltraZohm

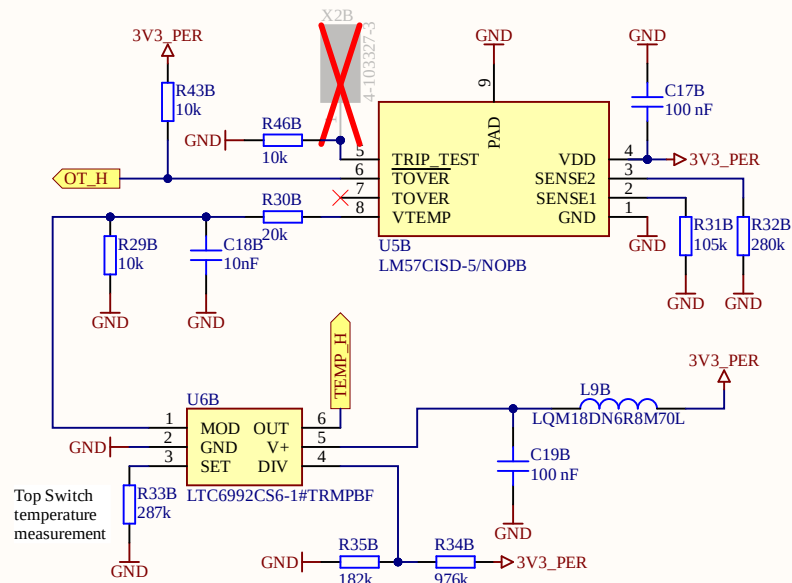
www.ultrazohm.com

Date: 07.03.2024

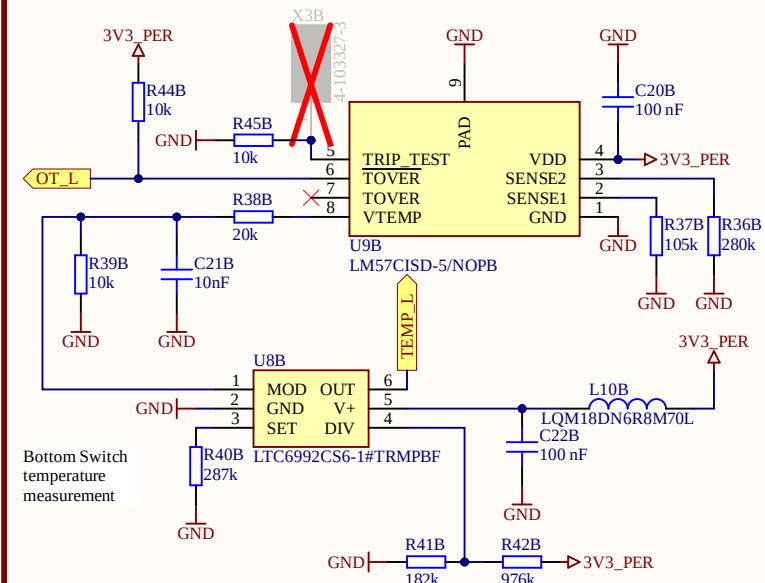
Sheet 8.1 of 14



Top MOSFET Temp Measurement

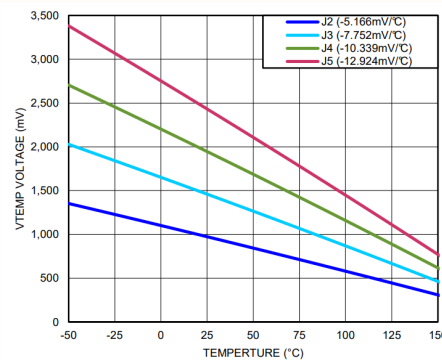


Bottom MOSFET Temp Measurement



$R_SENSE1 = \left(\frac{R31}{R37} \right) = 105k$
 $R_SENSE2 = \left(\frac{R32}{R36} \right) = 280k$
 is Gain
 -> J4 -> Leads to:
 1.11V@105°C
 2.00V@20°C

VTEMP has voltage divider with 20k/10k:
 MOD input voltages->
 $1.11V * 10 / (10 + 20) = 0.37V @ 105^\circ C$
 $2.00V * 10 / (10 + 20) = 0.67V @ 20^\circ C$
 MOD input should be between 0.14V and 0.9V



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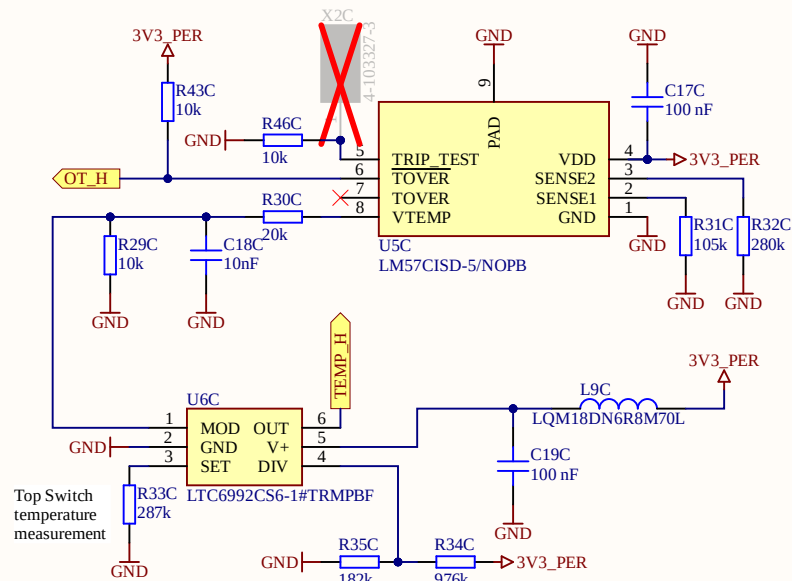
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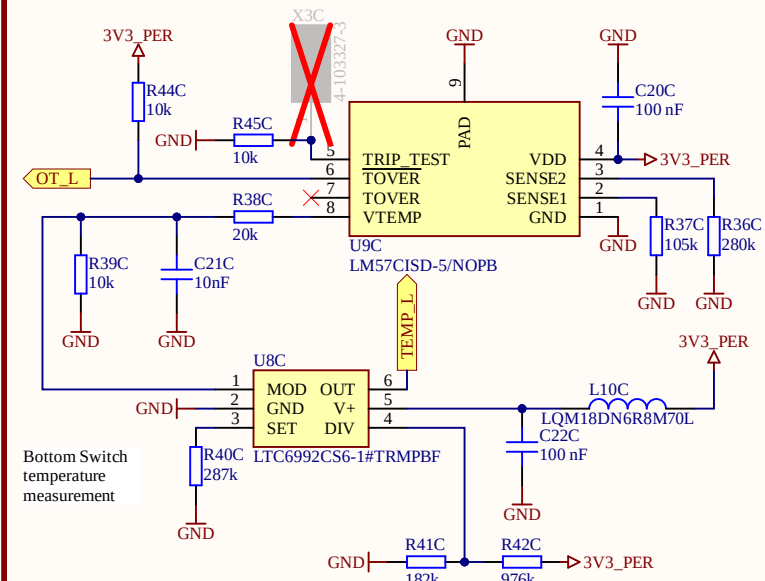
Sheet 8.2 of 14



Top MOSFET Temp Measurement



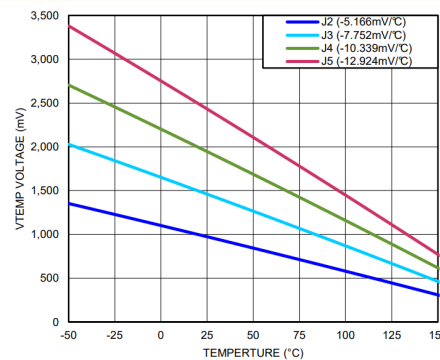
Bottom MOSFET Temp Measurement



$R_SENSE1 = \left(\frac{R31}{R37} \right) = 105k$
 $R_SENSE2 = \left(\frac{R32}{R36} \right) = 280k$
 is Gain
 -> J4 -> Leads to:
 1.11V@105°C
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VTEMP has voltage divider with 20k/10k:
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